

1. Work covered under this Clause of the Specification comprises the design and construction of foundations and other RCC constructions for switchyard structures, equipment supports, trenches, drains, control cubicles, bus supports, transformer, marshalling kiosks, auxiliary equipment & systems, buildings, tanks, boundary wall or for any other equipment or service and any other foundation required to complete the work. This clause is as well applicable to the other RCC constructions.

However, for the augmentation/bay extension works of existing substation, type of RCC structures and foundations etc. shall be similar to one already existing at the existing substation and for which design shall be furnished in support of safety of those RCC structures and foundations etc. Contractor must access the amount of work involved for the construction of switchyard structures, equipment supports, trenches, drains, control cubicles, bus supports, transformer, marshalling kiosks, auxiliary equipment & systems, buildings, tanks, boundary wall or for any other equipment or service and any other foundation required to complete the work for the existing substations.

2. Concrete shall conform to the requirements mentioned in IS: 456 and all the tests shall be conducted as per relevant Indian Standard Codes as mentioned in Standard field quality plan appended with the specification

A minimum grade for PCC and RCC shall be used for all structural/load-bearing members as per latest IS 456.

3. If the site is sloppy, the foundation height will be adjusted to maintain the exact level of the top of structures to compensate such slopes.

4. The switchyard foundation's plinths and building plinths shall be minimum 300mm and 500 mm above finished ground level respectively.

5. Minimum 75mm thick lean concrete shall be provided below all underground structures, foundations, trenches etc. to provide a base for construction.

6. Concrete made with Portland slag cement shall be carefully cured and special importance shall be given during the placing of concrete and removal of shuttering.

7. The design and detailing of foundations shall be done based on the approved soil data and sub-soil conditions as well as for all possible critical loads and the combinations thereof. The Spread footings foundation or pile foundation as may be required based on soil/sub-soil conditions and superimposed loads shall be provided.

8. If pile foundations are adopted, the same shall be cast-in-situ driven/bored or precast or under reamed type as per relevant parts of IS Code 2911. Only RCC piles shall be provided. Suitability of the adopted pile foundations shall be justified by way of full design calculations. Detailed design calculations shall be submitted by the bidder showing complete details of piles/pile groups proposed to be used. Necessary initial load test shall also be carried out by the bidder at their cost to establish the piles design capacity. Only after the design capacity of piles has been established, the Contractor shall take up the job of piling. Routine tests for the piles shall also be conducted. All the work (design & testing) shall be planned in such a way that these shall not cause any delay in project completion.

a. Design

1. Foundations shall be of reinforced cement concrete for new substation but for the augmentation / bay extension works of existing substation it could be of RCC/ PCC depending on type of structures and materials used for the similar type of structures in those bay extension works of existing substation. Design requirement shall be fulfilled by the Contractor and furnished for approval for both new substation and existing substation (for bay extension works) as specified in the scope of work. The design and construction of RCC/ PCC / Masonry structures shall be carried out as per IS: 456 and relevant IS code/CBIP manual/NBC etc and minimum grade of concrete shall be as per relevant IS code. Higher grade of concrete than specified above may be used at the discretion of Contractor without any additional financial implication to the Project Manager.

2. Limit state method of design shall be adopted unless specified otherwise in the specification.

3. For detailing of reinforcement IS: 2502 and SP: 34 shall be followed. Cold twisted deformed bars ($F_y=415 \text{ N/mm}^2$) conforming to IS: 1786 shall be used as reinforcement. However, in specific areas, mild steel (Grade I) conforming to IS: 432 can also be used. Two layers of reinforcement (on inner and outer face) shall be provided for wall & slab sections having thickness more than 150 mm. Clear cover to reinforcement towards the earth face shall be minimum 40 mm.

4. RCC water retaining structures like storage tanks, etc. shall be designed as un-cracked section in accordance with IS: 3370 (Part I to IV) by working stress method. However, water channels shall be designed as cracked section with limited steel stresses as per IS: 3370 (Part I to IV) by working stress method.

5. The procedure used for the design of the foundations shall be the most critical loading combination of the steel structure and or equipment and/or superstructure and other conditions, which produces the maximum stresses in the foundation or the foundation component and as per the relevant IS Codes of foundation design. Detailed design calculations shall be submitted by the bidder showing complete details of piles/pile groups or isolated /combined footings proposed to be used.

6. Design shall consider any sub-soil water pressure that may be encountered following relevant standard strictly.
7. Necessary protection to the foundation work, if required shall be provided to take care of any special requirements for aggressive alkaline soil, black cotton soil or any other type of soil which is detrimental/harmful to the concrete/masonry foundations.
8. RCC columns /pedestals shall be provided with rigid connection at the base.
9. All sub-structures shall be checked for sliding and overturning stability during both construction and operating conditions for various combinations of loads. Factors of safety for these cases shall be taken as mentioned in relevant IS Codes or as stipulated elsewhere in the Specifications. For checking against overturning, weight of soil vertically above footing shall be taken and inverted frustum of pyramid of earth on the foundation should not be considered.
10. Earth pressure for all underground structures shall be calculated using co-efficient of earth pressure at rest, co-efficient of active or passive earth pressure (whichever is applicable).
11. In addition to earth pressure and ground water pressure etc., a surcharge load of 1T/Sq.m shall also be considered for the design of all underground structures including channels, sumps, tanks, trenches, substructure of any underground hollow enclosure etc., for the vehicular traffic in the vicinity of the structure.
12. Following conditions shall be considered for the design of water tank in pumps house, channels, sumps, trenches and other underground structures:
 - i. Full water pressure from inside and no earth pressure & ground water pressure & surcharge pressure from outside (application only to structures, which are liable to be filled up with water or any other liquid).
 - ii. Full earth pressure, surcharge pressure and ground water pressure from outside and no water pressure from inside.
 - iii. Design shall also be checked against buoyancy due to the ground water during construction and maintenance stages. Minimum factor of safety of 1.5 against buoyancy shall be ensured ignoring the superimposed loadings.
13. The foundations shall be proportioned so that the estimated total and differential movements of the foundations are not greater than the movements that the structure or equipment is designed to accommodate.

14. The foundations of transformer and circuit breaker shall be of block type foundation. Minimum reinforcement shall be governed by IS: 2974 and IS: 456.

15. The tower and equipment foundations shall be checked for a factor of safety of 2.0 for normal condition and 1.5 for short circuit condition against sliding, overturning and pullout.

b. Admixtures & additives

1. Only approved admixtures shall be used in the concrete for the Works. When more than one admixture is to be used, each admixture shall be batched in its own batch and added to the mixing water separately before discharging into the mixer. Admixtures shall be delivered in suitably labeled containers to enable identification.

2. Admixtures in concrete shall conform to IS: 9103. The water proofing cement additives shall conform to IS: 2645. Project Manager shall approve concrete Admixtures/ Additives.

3. The Contractor may propose and the Project Manager may approve the use of a water-reducing set-retarding admixture in some of the concrete. The use of such an admixture will not be approved to overcome problems associated with inadequate concrete plant capacity or improperly planned placing operations and shall only be approved as an aid to overcoming unusual circumstances and placing conditions.

4. The water-reducing set-retarding admixture shall be an approved brand of Ligno-sulphonate type admixture.

5. The waterproofing cement additives shall be used as required / advised by the Project Manager.

c. Gates and Boundary Wall

1. The Gate frame shall be made of medium duty MS pipe conforming to relevant IS with welded joints.

2. The gates shall be fabricated with welded joints to achieve rigid connections. The gate frames shall be painted with one coat of approved steel primer and two coats of synthetic enamel paint.

3. Gates shall be fitted with approved quality iron hinges, latch and latch catch. Latch and latch catch shall be suitable for attachment and operation of pad lock from either side of gates. Hinges shall permit gates to swing through 180 degree back against fence. Gates shall be earthed by G I wire.

4. Gates shall be fitted with galvanized chain hook or gate hold back to hold gates open. Double gates shall be fitted with centre rest and drop bolt to secure gates in closed position.
5. Gates shall be installed in locations shown on drawings. Next to the main gate, a men gate (1.25 m wide, single leaf) shall also be provided.
6. Bottom of gates shall be set approximately 40mm above ground surface and necessary guiding mechanism shall be fitted.
7. The Contractor shall design and construct boundary wall around substation area as per requirements. The boundary wall shall be of height 2.0M and shall be made of RCC frame construction with RCC column and plinth beam arrangement and panels filled with one brick thick wall in cement sand mortar 1:6. The boundary wall shall be plastered on both external and internal faces with cement and sand plaster 1:6 of thickness 18 mm and 12 mm respectively. An additional barbed Y-shaped arm of MS angle 50x50x6 with 3-rows (6 nos) barbed wire A-4 IS: 278. Expansion joints shall be provided as per codal requirements. MS grating shall be provided at required locations for drainage purposes. The boundary wall shall be painted with minimum two coats of color wash over a base coat of white wash with lime. The front portion of boundary wall shall however be with a RC jail and 12 mm square MS bar top above brick work and pebble dash plaster finish with colour pigment. The steel work shall be given two coats of synthetic enamel paint of approved make over one coat of primer. Boundary wall and gate shall be constructed as per enclosed *tender drawing*.

11.00 Buildings - General Requirements

The scope for new control room building includes the design, engineering and construction including anti-termite treatment, plinth protection, DPC of Building including sanitary, water supply, electrification, false ceiling etc. of control room building. The buildings shall be of RCC framed structure of concrete of M20 grade (Min.). Following design criteria shall be adopted for design purposes for new substation.

If any extension of the Control Room building is required in augmentation / bay extension works of existing substation then extension part shall be compatible to existing one structurally and architecturally but following design criteria shall be adopted for design purposes for R&M of existing substation.

a. Control room Building

Minimum floor area requirements shall be 10000×12000 mm excluding space for wash room which may be increased at the time of detailed engineering to suit project requirements. The layout of the control room shall be finalised as per detailed engineering to suit project requirements. The minimum

dimension of different rooms required for C.R.building shall be as per drawing. The CR building shall consist of the following:

- a. Control room
- b. S/s In-charge room
- c. Battery room
- d. Store room
- e. Toilet

An open space of 1 m minimum shall be provided on the periphery of the rows of panel and equipment generally in order to allow easy operator movement and access as well as maintenance.

Any future possibility of annexe building shall be taken care of while finalizing the layout of the control room building.

Minimum headroom of 3 M below soffit of beams/false ceiling shall be considered for rooms. The roof shall have four side sloping roof or flat roof as finalised during detailed engineering.

i. Design

a) The buildings shall be designed:

1. To the requirements of the National Building Code of India, and the standards quoted therein.
2. For the specified climatic & loading conditions.
3. To adequately suit the requirements of the equipment and apparatus contained in the buildings and in all respects to be compatible with the intended use and occupancy.
4. With a functional and economical space arrangement.
5. For a life expectancy of structure, systems and components not less than that of the equipment, which is contained in the building, provided regular maintenance is carried out.
6. Be aesthetically pleasing. Different buildings shall show a uniformity and consistency in architectural design.
7. To allow for easy access to equipment and maintenance of the equipment.

8. With, wherever required, fire retarding materials for walls, ceilings and doors, which would prevent supporting or spreading of fire.
9. Suitable expansion joints shall be provided in the longitudinal direction wherever necessary with provision of twin columns.
10. Individual members of the buildings frame shall be designed for the worst combination of forces such as bending moment, axial force, shear force, torsion etc.
11. Permissible stresses for different load combinations shall be taken as per relevant IS Codes.
12. The building lighting shall be designed in accordance with the requirements of relevant section.
13. Sesmic considerations as applicable.

ii.Design loads

Building structures shall be designed for the most critical combinations of dead loads, super- imposed loads, equipment loads, wind loads, seismic loads, and temperature loads.

Dead loads shall include the weight of structures complete with finishes, fixtures and partitions and should be taken as per IS: 1911.

Super-imposed loads in different areas shall include live loads, minor equipment loads, cable trays, small pipe racks/hangers and erection, operation and maintenance loads. Equipment loads shall constitute, if applicable, all load of equipments to be supported on the building frame.

The wind loads shall be computed as per IS 875, Seismic Coefficient method shall be used for the seismic analysis as per IS 1893 with importance factor 1.5.

Wind and Seismic forces shall not be considered to act simultaneously.

Floors/slabs shall be designed to carry loads imposed by equipment, cables piping, equipment and other loads associated with building. Floors shall be designed for live loads as per relevant IS. Cable and piping loads shall also be considered additionally for floors where these loads are expected.

For consideration of loads on structures, IS: 875 shall strictly adhere to. Any other load coming in the structure, not mentioned in IS 875 shall be calculated as per relevant IS code and NBC.

iii. Submission

The following information shall be submitted for review and approval to the Project Manager:

1. Design criteria shall comprise the codes and standards used, applicable climatic data including wind loads, earthquake factors maximum and minimum temperatures applicable to the building locations, assumptions of dead and live loads, including equipment loads, impact factors, safety factors and other relevant information.
2. Structural design calculations and drawing (including construction/fabrication) for all reinforced concrete and structural steel structures.
3. Fully, dimensioned concept plan including floor plans, cross sections, longitudinal sections, elevations and perspective view of each building. These drawings shall be drawn at a scale not smaller than 1:75 and shall identify the major building components.
4. Fully dimensioned drawings showing details and sections drawn to scales of sufficient size to clearly show sizes and configuration of the building components and the relationship between them.
5. Product information of building components and materials, including walls partitions flooring ceiling, roofing, door and windows and building finishes.
6. A detailed schedule of building finishes including colour schemes.
7. A door & window schedule showing door types and locations, door lock sets and latch sets and other door hardware.

Approval of the above information shall be obtained before ordering materials or starting fabrication or construction as applicable.

iv. Finish Schedule

1. The finishing schedule is given in subsequent clauses.
2. M.S. Ladder should be provided to access the control room roof from outside. Ladder shall be made up of ISMC 75x40 which will run as beam one meter apart and intermediate steps will be made up of 45x45x5 angle with rise of 300 mm. Red oxide primer shall be applied initially, then two coats of rich zinc paint shall be applied to avoid corrosion.

v. Flooring

Flooring in various rooms of control room building shall be as for detailed schedules given in Table - 1

vi. Walls

Control room buildings shall be of framed superstructure. All walls shall be non-load bearing walls. Min. thickness of external walls shall be 230 mm (one brick) with 1:6 cement sand mortar.

vii. Plastering

All internal walls shall have minimum 12mm and 15 mm thick 1:6 cement sand plaster on either side of wall. The ceiling shall have 6mm thick 1:4 cement sand plaster.

viii. Finishing

All external surfaces shall have 18 mm cement plaster in two coats, under layer 12 mm thick cement plaster 1:5 and finished with a top layer 6 mm thick cement plaster 1:6 (DSR 13.19) with water proofing compound. The paint shall be antifungal quality of reputed brand suitable for masonry surfaces for high rainfall zone. White cement primer shall be used as per manufacturer's recommendation.

Internal finish Schedule is given Table - 1 below:

TABLE-1

S.No.	Location	Flooring & Skirting 150mm high	Wall Internal	Ceiling	Doors, Windows, Ventilators
1.	Control Room, Relay Room	Precast Terrazo tiles (DSR'02, item no. 11.29A.2 & 11.31.2)	Oil bound washable distemper on smooth surface applied with 2mm thick Plaster of Paris putty. (DSR'02 – 13.40 A & 13.77.2)	White Wash (DSR'02 – 13.70.1)	1) Standard steel rolled section frames with 5 mm glass. DSR'02 – 10.12, 10.13 & 10.14 2) Flush door shutters - DSR'02 – 9.25.2
2.	Sub-station Incharge,	Precast Terrazo tiles	Oil bound washable	White Wash	1) Standard steel rolled section frames with 5

	Office, corridor, staff room.	(DSR'02, item no. 11.29A.2 & 11.31.2	distemper on smooth surface applied with POP putty. (DSR'02 – 13.40 A & 13.77.2)	(DSR'02 – 13.70.1)	mm glass. DSR'02 – 10.12, 10.13 & 10.14 2) Flush door shutters - DSR'02 – 9.25.2
3.	Battery room	Acid and Alkali Resistant tiles. DSR'02 – 11.36 C. 1 & 11.36 C.1.1	Dado of acid resistant tile 1.2 M high & Paint above 1.2 M to ceiling. DSR'02 – 11.36 C. 2.1, 11.36C.2 & 13.96.1	Acid resistant Paint. DSR'02 – 13.96.1	1) Standard steel rolled section frames with 5 mm glass. DSR'02 – 10.12, 10.13 & 10.14 2) Flush door shutters - DSR'02 – 9.25.2 Painted with acid resistant Paint. DSR'02 – 13.96.1
4.	Toilet	Ceramic glazed tiles in flooring DSR'02 – 11.74	DADO glazed tile 2.1M high for toilet (DSR 02 - 11.73)	White Wash (DSR'02 – 13.70.1)	1) Standard steel rolled section frames with 5 mm glass. DSR'02 – 10.12, 10.13 & 10.14 2) Flush door shutters - DSR'02 – 9.25.2
5.	Other areas not specified	Terrazo tiles (DSR'02 - 11.29A.2 & 11.31.20	Oil bound distemper, DSR'02 - 13.40A & 13.77	White Wash (DSR'02 – 13.70.1)	

Note: DSR item references (DSR-2002) to be read with CPWD specifications are only for material and workmanship guidance of the Contractors.

ix.Roof

Roof of the C.R. Building shall consist of Cast-in-situ RCC slab treated with a water proofing system which shall be an integral cement based treatment conforming to CPWD specification (item no. 25.8 of DSR 2002). The water proofing treatment shall be of following operations:

- i. Applying and grouting a slurry coat of neat cement using 2.75 kg/m^2 of cement admixed with proprietary water proofing compounds conforming to IS: 2645 over the RCC slab including cleaning the surface before treatment.
- ii. Laying cement concrete using broken bricks/brick bats 25mm to 100mm size with 50% of cement mortar 1:5 (1 cement: 5 coarse sand) admixed with proprietary water proofing compound conforming to IS: 2645 over 20mm thick layer of cement mortar of min 1:5 (Cement: 5 coarse sand) admixed with proprietary water proofing compound conforming to IS: 2645 to required slope and treating similarly the adjoining walls upto 300mm height including rounding of junctions of walls and slabs.
- iii. After two days of proper curing applying a second coat of cement slurry admixed with proprietary water proofing compound conforming to IS: 2645.
- iv. Finishing the surface with 20mm thick joint less cement mortar of mix 1:4 (1 cement: 4 coarse sand) admixed with proprietary water proofing compound conforming to IS: 2645 and finally finishing the surface with trowel with neat cement slurry and making of 300 x 300 mm square.
- v. The whole terrace so finished shall be flooded with water for a minimum period of two weeks for curing and for final test. All above operations to be done in order and as directed and specified by the Project Manager.

With average thickness of 120 mm and minimum thickness at khurra at 65 mm.

x. Glazing

Minimum thickness of glazing shall be 5.0 mm. as per IS: 2835.

xi. Doors and Windows

The details of doors and windows of the control room building shall be as per finish schedule Table-1 and tender drawing with the relevant IS code. Rolling steel shutters and rolling steel grills shall be provided as per layout and requirement of buildings. Paints used in the work shall be of best quality specified in CPWD specification.

xii. Plumbing & Sanitation

1. All plumbing and sanitation shall be executed to comply with the requirements of the appropriate byelaws, rules and regulations of the Local Authority having jurisdiction over such matters. The

Contractor shall arrange for all necessary formalities to be met in regard to inspection, testing, obtaining approval and giving notices etc.

2. PVC syntax or equivalent make Roof water tank of adequate capacity depending on the number of users for 24 hours storage shall be provided. Minimum 1 Nos. 500 liters capacity shall be provided.

3. Galvanized MS pipe of medium class conforming to IS: 1239 shall be used for internal & external piping work for potable water supply.

4. Sand CI pipes with lead joints conforming to IS: 1729 shall be used for sanitary works above ground level.

5. Each toilet shall have the following minimum fittings.

(a) WC (Western type) 390 mm high with toilet paper roll holder and all fittings

Or

WC (Indian Type) Orissa Pattern (580 x 440 mm) with all fittings (both types of WCs shall be provided at alternate locations).

(b) Urinal (430 x 260 x 350 mm size) with all fittings.

(c) Wash basin (550 x 400 mm) with all fittings.

(d) Bathroom mirror (600 x 450 x 6 mm thick) hard board backing

(e) CP brass towel rail (600 x 20 mm) with C.P. brass brackets

(f) Soap holder and liquid soap dispenser.

6. All fittings, fastener, grating shall be chromium plated.

7. All sanitary fixtures and fittings shall be of approved quality and type manufactured by well known manufacturers. All items brought to site must bear identification marks of the type of the Manufacturer.

8. Soil, waste and drain pipes, for underground works shall be stoneware for areas not subject to traffic load. Heavy-duty cast iron pipes shall be used otherwise.

9. In case of Augmentation/R&M of existing substation, amount of work shall be envisaged by contract for lump sum quotation.

12.00 Miscellaneous General Requirements

1. Dense concrete with controlled water cement ratio as per IS-code shall be used for all underground concrete structures such as pump-house, tanks, water retaining structures, cable and pipe trenches etc. for achieving water-tightness.
2. All joints including construction and expansion joints for the water retaining structures shall be made water tight by using PVC ribbed water stops with central bulb. However, kicker type (externally placed) PVC water stops shall be used for the base slab and in other areas where it is required to facilitate concreting. The minimum thickness of PVC water stops shall be 5 mm and minimum width shall be 230 mm.
3. All steel sections and fabricated structures that are required to be transported on sea shall be provided with anti corrosive paint to take care of sea worthiness.
4. All mild steel parts used in the water retaining structures shall be hot-double dip galvanised. The minimum coating of the zinc shall be 750 gm/sq. m. for galvanised structures and shall comply with IS: 2629 and IS: 2633. Galvanizing shall be checked and tested in accordance with IS: 2633. The galvanizing shall be followed by the application of an etching primer and dipping in black bitumen in accordance with BS: 3416.
5. A screed concrete layer not less than 100 mm thick and of grade not weaker than M10 conforming to IS: 456-1978 shall be provided below all water retaining structures. A sliding layer of bitumen paper or craft paper shall be provided over the screed layer to destroy the bond between the screed and the base slab concrete of the water retaining structures.
6. Bricks having minimum 75kg/cm² compressive strength can only be used for masonry work. Contractor shall ascertain himself at site regarding the availability of bricks of minimum 75kg/cm² compressive strength before submitting his offer.
7. Doors and windows on external walls of the buildings (other than areas provided, with insulated metal claddings) shall be provided with RCC sunshade over the openings with 300 mm projection on either side of the openings. Projection of sunshade from the wall shall be minimum 450 mm over window openings and 750 mm over door openings.
8. Service ladder shall be provided for access to all roofs.

9. Angles 45x45x5 mm (minimum) with lugs shall be provided for edge protection all round cut outs/openings in floor slab, edges of drains supporting grating covers, edges of RCC cable/pipe trenches supporting covers, edges of manholes supporting covers, supporting edges of manhole precast cover and any other place where breakage of corners of concrete is expected.

10. Anti termite chemical treatment shall be given to column pits, wall trenches, foundations of buildings, filling below the floors etc. as per IS: 6313 and other relevant Indian Standards.

11. All rungs for ladder shall also be galvanised as per IS: 277 medium classes.

12. For all civil works covered under this specification, nominal mix by volume batching as per CPWD specification is intended. The relationship of grade of concrete and ratio of ingredients shall be as below:

Sl.No.	Mix	Cement	Sand	Coarse aggregate of 20 mm down grade as per IS 383
1.	M 10	1	3	6
2.	M 15	1	2	4
3.	M 20	1	1.5	3

The material specification, workmanship and acceptance criteria shall be as per relevant clauses of CPWD specification and approved standard Field Quality Plan.

13. The details given in tender drawings shall be considered along with details available in this section of the specification while deciding various components of the building.

14. Items/components of buildings not explicitly covered in the specification but required for completion of the project shall be deemed to be included in the scope.

13.00 Interfacing

The proper coordination & execution of all interfacing civil works activities like fixing of conduits in roofs/walls/floors, fixing of foundation bolts, fixing of lighting fixtures, fixing of supports/embedment, provision of cutouts etc. shall be the sole responsibility of the Contractor. He shall plan all such activities in advance and execute in such a manner that interfacing activities do not become bottlenecks and dismantling, breakage etc. is reduced to minimum.

14.00 Water Supply

- (i) Contractor shall make its own arrangement for construction water.
- (ii) The Contractor shall carry out all the plumbing/erection works required for supply of water in control room building.
- (iii) The details of tanks, pipes, fittings, fixtures etc for water supply are given elsewhere in the specification under respective sections.
- (iv) A scheme shall be prepared by the Contractor indicating the layout and details of water supply which shall be got approved by the Project Manager before actual start of work including all other incidental items not shown or specified but as may be required for complete performance of the works.
- (v) Bore wells and pumps for water supply are in the scope of Contractor meeting the day-to-day requirement of the water supply.
- (vi) If the water is supplied by Municipal Corporation then bore well for water supply purposes is not required to be carried out by Contractor. Contractor shall also make necessary arrangement /formalities to receive water connection from corporation.

15.00 Sewerage System

- (i) Sewerage system shall be provided for control room building.
- (ii) The Contractor shall construct septic tank and soak pit suitable for 5 users if outside of Municipal Corporation zone. Otherwise, all necessary arrangement for the disposal of sewerage to the Municipal Corporation's end shall be arranged by the Contractor at his own cost for regularizing the disposal activity.

(iii) The septic tank and soak pit shall be constructed as per enclosed tender drawing.

16.00 Statutory Rules

a. Contractor shall comply with all the applicable statutory rules pertaining to factories act (as applicable for the State). Fire Safety Rules of Tariff Advisory Committee, Water Act for pollution control etc.

b. Provisions for fire proof doors, no. of staircases, fire separation wall, plastering on structural members (in fire prone areas) etc. shall be made according to the recommendations of Tariff Advisory Committee.

c. Statutory clearance and norms of State Pollution Control Board shall be followed as per Water Act for effluent quality from plant.

d. Requirement of sulphate resistant cement (SRC) for sub structural works shall be decided in accordance with the Indian Standards based on the findings of the detailed soil investigation to be carried out by the Bidder.

e. Foundation system adopted by Bidder shall ensure that relative settlement and other criteria shall be as per provision in IS: 1904 and other Indian Standards

f. All water retaining structures designed as un-cracked section shall also be tested for water tightness at full water level in accordance with clause no. 10 of IS: 3370 (Part-I).

g. Construction joints shall be as per IS: 456.

h. All underground concrete structures like water retaining structures etc. shall have plasticizer cum water proofing cement additive conforming to IS: 9103. In addition, limit on permeability as given in IS: 2645 shall also be met with. The concrete surface of these structures in contact with earth shall also be provided with two coat of bituminous painting for water/damp proofing. In case of water leakage in the above structures, Injection Method shall be applied for repairing the leakage.

i. All building/construction materials shall conform to the best quality specified in CPWD specifications if not otherwise mentioned in this specification.

j. All tests as required in the standard field quality plans have to be carried out.

17.00 Fencing

a. Product materials for fencing

The minimum requirements are as follows:

i.Chain Link fence fabric in accordance to IS-2721

1. Size of mesh : 75 mm
2. Nominal wire size : 4.0 mm dia
3. Width of chain link : 2000 mm
4. Class of zinc coating : medium
5. Zinc coated after weaving.

ii.Posts

Angle Section

Intermediate : L 65 x 65 x 6

Straining posts : L 65 x 65 x 6

Stay post : L 65 x 65 x 6

1. All structural steel shall conform to IS: 2062 and shall be painted with a coat of approved steel primer and two coats of synthetic enamel paint.
2. The Chain Link fabric shall be fixed to the post at the top and bottom of the fence by welding/fixing 50 mm MS flat all through its length.
3. Fencing top shall be either of galvanised barbed wire or tape. Barbed wire shall conform to IS: 278.
4. The barbed wire may consist of not more than two splices per reel. The barbed wire shall be formed by twisting two line wires, one containing the barbs. The barbed wire shall be designated as A-4 IS: 278 and shall be galvanized.

5. Above chain link, 3-rows (6 nos) of barbed tape/wire shall be provided in each arm of the Y shaped barbed arm at top.
6. With barbed tape/wire above the chain link fence, the total fence height shall be minimum 2500 mm above finished gravel level.
7. Barbed tape/wire arms shall be same as intermediate and straining post.
8. Tension wire: single strand, high tensile, galvanised steel wire, 4 mm diameter.
9. Fittings and hardware: cast aluminum alloy or galvanized steel, malleable or ductile cast iron turnbuckles to be drop forged.
10. GI chain link mesh shall be as per IS: 2721. Mesh size 75 mm and nominal wire size shall be 4.0 mm diameter.

On the results of these additional tests, the whole or portion of the barbed wire/tape shall be accepted or discarded by the Purchaser, as the case may be.

b. Installation

1. Contractor shall submit the fencing drawing Fence shall be installed along lines shown on approved drawings.
2. Post holes shall be excavated by approved methods.
3. Intermediate posts shall be spaced 2.5 m apart measured parallel to ground surface.
4. Straining posts shall be installed at equal intervals not exceeding 25.0 m.
5. Straining posts shall be installed at sharp changes in grade, at corners, at change of direction and where directed.
6. All corner post will have two-stay post and every tenth post will have a transverse stay post.
7. Posts shall be set in 1:2:4 plain cement concrete Blocks of minimum dimension 400 mm x 400 mm x 1000 mm deep Concrete work shall conform to relevant clause. Post shall be braced and held in plumb position and true alignment and elevation until concrete has set.

8. Fence fabric shall not be installed until concrete has cured a minimum of 7 days.
9. Bottom and top of the fence fabric shall be fixed with MS flats of 50 mm x 6mm (min).
10. Fence fabric shall be laid out with barbed edge on top, stretched tightly and shall be fastened to intermediate, post gate and straining post with 50 x 6 flats.
11. Fabric shall be secured to tension wires with tie wires at 400 mm intervals. Tie wires shall be given not less than two twists.
12. Barbed tape shall be spliced with standard wire splices.
13. Barbed tape shall be stretched to have uniform tension.
14. Barbed tape shall be attached to barbed wire arms with approved metal clips.
15. Toe wall of one Brick/Random Rubble masonry, with notches over 75 mm thick PCC (1:4:8) shall be provided below all fencing and shall be minimum 200 mm above and 200 mm below finished ground level. All exposed surfaces of brick toe wall shall be provided with 1:6 cement sand plaster and coated with two coats of colour wash with a base coat of white wash with lime. Rubble masonry toe wall shall be with raised & cut pointing and 50 mm PCC (1:2:4) band coping.
16. Proper earthing shall be done for fencing also.

13.9.LT AB Cable Reconductoring Work

1. Survey

1.1. Survey

The Contractor shall carry out a GPS based survey of existing bare conductor LT distribution lines in the habitation. The Survey should cover Pole by pole survey of all the bare conductor lines to identify the location of poles and phase configuration, pole condition, Existing stays / struts, Existing street lamp connections, Location and capacity of the connected distribution transformer (DTR) to each LT line, Connection points of any existing ABC lines connected to the bare conductor lines should be identified. The length of such ABC lines and total service connections provided through this line should be indicated on the map at this point, Load readings for each LT feeder. Upon completion of the survey the Contractor is required to develop the Single Line Diagram (SLD-A) indicating the survey Information and Schedule of network survey information (Schedule – A).

1.2. Load Readings

Load readings of each LT feeder shall be taken before and after proposed ABC conversions. This is required to monitor actual demand reductions achieved following ABC conversions.

Load currents on each phase and the neutral in respective LT feeder shall be taken on four different occasions per day before and after completion of ABC conversion work. The load current measurements shall be decided by Engineer in Charge. Load reading before ABC conversions shall be taken when carrying out the survey and recorded in Schedule-A. Load readings after ABC conversions shall be provided to the Project Manager/Engineer in charge within one week from completion of works.

1.3. Network Design for ABC Conversions

Contractor should prepare details of proposed ABC conversions for respective habitation in accordance with these guidelines and prepare Single line diagram indicating proposed ABC conversions (SLD-B) and Schedule of proposed ABC conversions (Schedule -B). while preparing SLD, loading in distribution transformer must be examined. On completion of work, average day-loading in distribution transformer should limit 80-85% only. Contractor

1.4. Load Balancing

When preparing ABC conversion schedule (Schedule-B), phase connections for single phase Distribution Box/SMC Distribution Boxes shall be determined so that total number of single phase consumers are balanced across the three phases of a given feeder.

2. Conversion of bare conductor lines to ABC

2.1. AB Cable types and Sizes

1,1 kV voltage grade XLPE insulated aluminum conductor and aluminum alloy bare neutral messenger type cables shall be used for proposed ABC conversions. The AB cables provided shall fully comply with technical specifications provided in this tender document.

The following **standard sizes** of AB cables shall be used:

- ABC16-SP : 1X16 mm² (ph)+1X25 mm² (bare messenger cum neutral) +1x16 mm² (insulated street lighting cable)
- ABC50 : 3X50 mm² (ph)+1x35 mm² (bare messenger cum neutral) +1x16 mm² (insulated street lighting)
- ABC95 : 3X95 mm² (ph)+1x70 mm² (bare messenger cum neutral) +1x16 mm² (insulated street lighting)

2.2. Vertical and Horizontal Clearances

all statutory clearances shall be ensured for ground clearance, line-to-line clearance, road crossing clearance, horizontal and vertical clearances from buildings/objects etc. All road crossings and line crossings shall be guarded as per specifications. Conductor joint should not be provided in mid span length. Instead, it should be nearer to the support.

As per ISS 162-1961 minimum electrical clearance from live part to earth and safety clearance in case of different voltage must be kept as follows:			
Voltage	Electrical Clearance (mm)		Safety Clearance in SIS (mm)
	Phase - Earth	Phase - Phase	
33kV	381	432	2740
66kV	658	786	3050

Minimum Clearance Between Power Lines (mtr.) :			
kV	11	33	66
11	2.44	2.44	2.44
33		2.44	2.44
66			2.44

2.3. Installation of AB Cables

- Prior to installation of AB cables, all pole works including **stay/struts works** should be completed as per scope of works provided in proceeding sections. (i.e installation of new poles, pole replacements, pole re-alignment, installation of pole supports).

All ABC accessories used for installation works shall conform to technical specification provided in this document.

3.1. Installation of clamp assemblies

AB cable should be installed on poles using anchoring and suspension clamps according to the approved drawings by the Project Manager. Samples of complete clamp assemblies shall be approved by the Project Manager prior to use.

Anchoring clamps shall be used at the beginning and end of each cable run, at a major change in direction, terminal poles and at T-off points. Suspension clamps shall be used at other intermediate points.

It should be noted that different clamps are specified for cable ranges 25-50 sqmm and 70-95 sqmm. These have different dimensional and maximum load specifications. If ABC manufacturer recommends any alternate clamps it should be approved by the Project Manager prior to use.

Stainless steel straps and buckles shall be used for fixing pole brackets to the pole as shown in drawings. Strap binding tool shall be used for tensioning and cutting the straps.

Separation of neutral messenger for tensioning and fixing to the clamp should be done using plastic phase separators. Weather resistant black nylon ties should be used for tying insulated conductors to the neutral messenger at either side of suspension clamps, to prevent the phase conductors from chafing against suspension clamp.

3.2. Stringing of AB cable

Stringing of AB cables shall be done in a proper manner ensuring insulated conductors do not get damaged during installation. Dragging the ABC on the ground is not permitted. Pulleys installed on poles shall be used to pull AB cables.

Minimum clearance above ground to line shall be maintained. Sag tension charts for installing AB cables shall be developed by the Contractor taking into consideration of cable characteristics, maximum / minimum temperatures and maximum wind pressure as per service conditions provided by Employer. Based on this clear guidelines shall be provided to linesmen to ensure bare neutral messenger is pulled at appropriate tension so that;

- (a) Required ground clearances are maintained, and
- (b) Messenger conductor tension is maintained well below its breaking load at all temperatures.

Dynamometer method or sag method may be used to ensure appropriate tension of neutral messenger during installation. Over tensioning of neutral messenger should be avoided to ensure its tension does not exceed permissible loading limits at low temperatures. Loose spans of AB cable should be avoided to maintain permissible maximum sag at high temperatures. Loose spans may only be allowed for short spans in special cases. This applies when there are practical difficulties to install necessary stays or struts as required at a t-off point. A short loose span of AB cable may be used in this case to transfer the stay / strut support point upstream or downstream of the line.

Stringing of AB cable shall be done using proper equipment such as stringing blocks with plastic coated pulleys, pulling (come along) clamp, cable hoist and pulling tool, dynamometer etc. Proper

equipment recommended by the ABC manufacturer shall be used to avoid any damage to the cable during installation. Temporary stays or strut poles shall be employed as necessary during stringing operation to ensure safety of personnel and equipment.

Phasing of insulated conductors shall be identified by one, two and three ridges on the XLPE insulation. Same phasing shall be maintained accordingly through the line. Interchanging of phasing at any connection point is not permitted.

In order to ensure durability of AB cables and to prevent possibility of failures due to effects of water treeing, any exposed parts of phase conductors or open cuts of insulation are strictly not permitted. Hence all connectors to be used in ABC line shall be pre-insulated type or bare connectors covered by heat shrinkable tubing or GelWrap sleeves. For installing connectors proper equipment such as insulation stripping tool, ratchet cable cutter, hydraulic compression tool with compression dies shall be used.

All cable ends shall be properly sealed by pre-moulded or heat-shrinkable type end caps.

Samples of all ABC accessories including connectors shall be approved by the Project Manager prior to use.

Mid-span joints shall be generally avoided by properly planning stringing work. In exceptional cases where mid span joint is required, pre-insulated compression connectors shall be used. The joints for each phase shall be staggered along the cable. No mid-span joints are allowed for AB cable sections running across a street.

Insulated piercing connectors or non-tension mechanical connectors with heat shrinkable tubing shall be used for non-tension inline connections at anchoring points where necessary.

3.3. AB cable connections to Distribution Transformers

AB cables shall be connected to bus-bars or protection equipment terminals of the LT feeder Distribution Boards by means of pre-insulated compression lugs and aluminium / bi-metallic strips.

Where no Distribution Board is available AB cable shall be directly connected to the distribution transformer bushing terminals using insulated compression lugs. In this case additional length of AB cable shall be provided by means of a loop to facilitate future connection to the Distribution Board or fuses.

3.4. Earthing of Poles / ABC neutral messenger conductor

Earthing shall generally be carried out in accordance with the requirements of latest CEA regulations (as amended from time to time) and the relevant regulations of the Electricity Supply Authority

The Contractor shall ensure every 6th pole of ABC line including neutral messenger and any metallic hardware is earthed with spike earth (20x2500 mm) as per existing practice of Employer. (for normal soil).

Poles shall be earthed using 8 SWG (7/4.0 mm) GI wire with 1 No. Coil/Spike/Pipe earth.

3.5. Pole Numbering

Each pole of the existing line shall be uniquely numbered as per pole numbering scheme followed by the Employer. Pole number and other information as required by the Employer should be painted on the pole.

3.6. Dismantling existing bare conductors and line hardware

Dismantling existing bare conductors and line hardware. Conductors and other line hardware including insulators, brackets, cross arms and bolts and nuts shall be carefully removed without causing damage to the existing poles. Bare conductors shall be removed in the longest length practicable for future re-use with a metal tag of description/ tag # of conductor, the said conductor shall be wound on empty conductor reels or made up in rolls.

Following dismantling works affected areas shall be cleaned and reinstated. All dismantled items shall remain the property of Employer and Contractor shall deliver all salvaged materials to the designated Employer warehouse as directed by the Project Manager.

3.7. Parallel AB Cable lines

Parallel AB cable lines refers to installation of second ABC cable line on the existing poles as per design requirements approved by Employer.

The scope of work for this item shall exclude (a) existing bare conductor line dismantling and (b) pole numbering requirements applicable for the scope of works specified above for 'conversion of bare conductor lines to ABC'.

3.8. Rates for Conversion of bare conductor lines to ABC

It should be noted that scope of work for this item relates to AB cable installation on existing poles. It excludes any additional works required on the existing poles or installation of new poles, installation of Distribution Box/SMC Distribution Boxes, stays etc. Separate BOQ items are provided for work associated with installation of new poles and replacement, re-alignment or relocation of existing poles, installation of stays and struts and installation of Distribution Box/SMC Distribution Boxes for AB cables.

4. Installation of LT Distribution Box/SMC Distribution Boxes for ABC

4.1. Types of LT Distribution Box/SMC Distribution Boxes

The following types of LT Distribution Box/SMC Distribution Boxes (according to number of consumers to be connected) shall be selected:

- Type A : Single Phase, 1 Incoming (25 mm²)/ 6 outgoing (upto 10 mm²)
- Type B : Three Phase, 1 Incoming (35 mm²)/ 4 outgoing (upto 16 mm²)

4.2. Mounting Arrangement

Distribution Box/SMC Distribution Box (DB) shall be mounted on LT pole with galvanized MS clamp of 40x3 mm size.

4.3. Connection to AB Cable

2Cx25 sqmm or 4Cx35 sqmm Stranded cables shall be used to connect AB cable with single phase and three phase Distribution Box/SMC Distribution Box respectively. For connection to ABC, insulation piercing connectors (IPC) and PG clamps of appropriate size shall be used.

Single phase Distribution Box/SMC Distribution Boxes shall be connected to specified phase as per Schedule-B in order to ensure load balancing in ABC line.

4.4. Connection of Consumer Service Cables

All existing consumer service cables shall be re-connected to the Distribution Box/SMC Distribution Box by the Contractor. In case existing consumer service connections are to be replaced with armoured service cables in a given habitation as determined by the Project Manager, new armoured cable shall be connected to the Distribution Box/SMC Distribution Box.

5. Replacement of Existing Consumer Service Connections

5.1. Service Connection Types

The applicable service connection types are as follows;

SC Type	Connected Load No of Phases / Contract Dmd	Service Cable Size (cores / sqmm)
SP-1	Single Phase / upto 4 kW	2 x 4 mm ²
TP-1	Three Phase / upto 4 kW	4 x 4 mm ²
TP-2	Three Phase / above 4 kW	4 x 4 mm ²

5.2. Service Cable Types

The service cable shall be 1.1 KV grade PVC insulated, PVC sheathed, armoured multicore **stranded** aluminium cable as per sizes indicated in above table

5.3. Service Cable Span

Permissible maximum span for service cable shall be 30 m. In isolated cases this limit may be extended with the approval of Project Manager provided that required ground clearance is maintained with additional supports where necessary.

Existing service cables shall be replaced with armored service cables only in certain specified areas which are high theft prone areas. This will be specified by the Project Manger during

execution. No service cable replacements shall be carried out without the approval of Project Manager.

In this case existing service cables shall be replaced with new armoured cables. Existing un-armoured service cables shall be removed and returned to Employer warehouse. Any non-standard supports used as supports for existing service cables shall also be removed.

New armoured service cables shall be drawn from the LT Distribution Box/SMC Distribution Box upto the meter board as shown in drawings. The service wire is to be hanged on supportive GI wire between pole support and the house. 7/3.15 mm (10 SWG) & 7/4.00 mm (8 SWG) GI wires shall be used for single phase and three phase services respectively.

Before installing service wires and GI wire, GI pipe / MS Angle on the consumer premises is to be erected using clamps/ nails/proper binding etc. In case of hut or poor structure at consumer premises, GI pipe is to clamp on wooden planks/wooden structure existing in the house. The GI pipe should be supported for neutralizing tension by means of GI tie wire support. In pukka/brickwork/cement concrete foundations, house, GI support pipe is to be clamped by means of MS clips.

New service cable shall be connected to existing consumer meter terminals or incoming fuse / MCB / MCCB terminals. Care should be taken not to damage the existing meters in the process. 20mm dia PVC conduit pipes shall be used to take down service cable from the roof to the meter box along the walls. The service cable shall be drawn inside PVC pipes from roof upto the meter board.

Terminal cover of the meter or fuses shall be sealed upon connection of new service cable as arranged with the Employer.

Earth terminal point shall be provided at meter board via bearer GI wire. This point shall be connected with the proper earthing system through GI wire. 10mm diameter earth knob in form of bolt and nut is to be installed on energy meter board.

All work required to install service cable from LT Distribution Box/SMC Distribution Box upto the meter board shall be carried out as per drawings. This includes GI wire for supporting service cables from LT Distribution Box/SMC Distribution Box upto the consumer premises, MS angle fixed at roof, clamp with bolts nuts and flat iron for fixing GI wire on pole, clamps for fixing cable to GI wire, PVC conduit pipes and accessories to draw the service cable from roof upto the meter board.

Contractor

6. Installation of Poles

6.1. Types of pole installation works for ABC conversions

The following types of pole installation works applicable for ABC conversions as per site requirements specified in the table below;

#	Type	Site Requirements
1	Installation of new poles	• New pole to be used as an intermediate pole in the existing line to address excessive spans or other requirements. • When there is a need to do minor extensions to the existing line in order to minimize excessive service cable lengths to existing consumers. • To shift existing lines due to safety/clearance issues.
2	Replacement of existing (unusable) poles	• To replace existing poles which are damaged / corroded and cannot be re-used. • To replace of existing non-standard line supports such as temporary structures including supports embedded in buildings.
3	Re-location of existing poles	• To shift existing poles • To shift existing lines due to safety/clearance issues
4	Re-alignment of existing poles	• To straighten the poles that are inclined in a particular direction but in re-usable condition

6.2. Types of poles

For works relating to new poles and replacement of poles, poles according to Employer specification shall be used.

For works relating to re-location or realignment of existing poles the available existing pole (any type) may be re-used provided that it is in re-usable condition.

7. Erection of Poles

7.1. Removal of existing poles for re-use or disposal

All unusable poles or non-standard line supports in the existing LT lines shall be removed and disposed. If the existing pole is removed for re-use adequate measures should be taken not to damage the pole during removal.

Existing poles shall be removed by pulling the complete pole from the ground; poles shall not be cut off at the ground line. Pole shall be cleaned and any material attached with the pole (including concrete) shall be removed. The RCC base plate may be removed and re-used if it is in re-usable condition.

Pits shall be backfilled and compacted completely with sufficient added backfill piled above grade to prevent depressions being created by natural compaction.

Contractor shall be responsible for disposal of unusable poles or non-standard line support structures after taking approval of the Project Manager/ Engineer In- Charge of the Employer.

7.2. Erection of new or existing poles

For 8/8.5 m PCC poles pole pit shall be excavated as per details provided in drawing # <REC/DDUGJY/GEN/02. Pole shall be placed on the RCC base pad made as per drawing # REC/DDUGJY/GEN/05A. PCC pole pit shall be refilled with 200 mm average size of boulder mixed with excavated earth. Proper ramming shall be performed for better compaction. Pole shall be erected fully vertical and firmly fixed to ground and shall not wobble.

7.3. Erection of poles with concrete foundation

Concrete foundations shall be used to erect terminal / tension poles and poles in water logged areas or for all locations as instructed by the Project Manager. Details of concrete foundation for PCC poles are provided in drawing # REC/DDUGJY/GEN/01.

7.4. Re-alignment of existing pole

Existing poles that are inclined in a particular direction shall be re-aligned / straightened by pulling, providing additional bouldering, concreting and re-compacting as necessary. Upon re-alignment pole shall be erected fully vertical and firmly fixed to ground and shall not wobble. If necessary pole may be completely removed and re-installed.

7.5. Earthing of poles

Earthing of poles shall be carried out as per CEA regulations and existing practice of Employer.

7.6. Pole Numbering

Requirements specified in section 3.9 with respect pole numbering shall also be applicable for new poles or pole replacements indicated in this section.

8. Installation of Stays and Struts

8.1. Installation of Stays (Guys)

Stays shall be installed to nullify tension on poles due to tension of AB cable at terminal, angle, cut-point and T-off positions. Stays may also be installed at steep gradient locations as required. Along the straight run stays shall be installed at minimum two locations in 1 km.

If there are no existing stays installed at such locations in the existing line, such poles shall be identified and included in Schedule-B for installation of new stays. In addition required stays for proposed new and relocated poles shall also be included.

10 SWG stay wire (7/ 3.15 mm) with 16 mm stay rod shall be used for 11 kV / LT lines. Stay shall be installed in the opposite direction of resultant force due to AB cable tension in order to nullify the same.

If the stay wire proves to be hazardous, it should be protected with suitable asbestos pipe filled with concrete of about 2 m length above the ground level, painted with white and black strips so that, it may be visible at night.

8.2. **Installation of Struts**

The struts may be used only in case where stays cannot be installed due to physical obstacles or limitations. Strut shall be applied in the same direction of resultant force due to AB cable tension in order to nullify the same.

8.3. **Installation of Stays**

Stay assembly comprising of turn buckle assembly, anchor rod and plate, stay insulator, thimble and GI stay wire shall be supplied and installed as indicated in drawing # EC/DDUGJY/GEN/09B.

In general, the stay should be applied on the pole as close as possible to the load center. The angle between stay wire and pole shall be between 45° – 60°. Where there are issues obtaining specified angle due to physical obstacles, bow (outrigger) stay arrangement may be considered with the approval of the Project Manager. In case of critical space issues, fly stays may also be considered subject to approval of the Project Manager. (Reference : REC Drawing # G4).

Concreting of stay pit shall be done as shown in drawings except for firm soil where compacting with necessary aggregates shall be done.

Stay wire shall be properly tightened after installation and allowing sufficient time for setting concrete. Contractor shall ensure all stays in the existing line are properly tightened including new and existing stays.

8.4. **Installation of Strut pole**

Installation of strut pole shall be carried out using 8 m PCC pole and pole brackets as shown in drawing # REC/DDUGJY/GEN/10. Strut pole shall be installed with the RCC base plate. Angle between the line and strut pole shall be 45°.

9. **Re-Connecting Existing Street Lamps to ABC**

If there are existing street lamps connected to the bare conductor line, these street lamps shall be re-connected to ABC line through the street lighting conductor using IPC (for insulated conductor) and PG clamps (for bare neutral messenger). This work scope is limited to providing connection to existing street lamps only.

10. **Shutdown during execution of works :**

The Contractor is required to take shutdown to execute reconductering and all other works wherever needed. During shut down, safety of system and operating manpower shall be ensured by Contractor.

Shut down shall be planned with concerned substation incharge well in advance. This may subject to exigencies leading to cancellation of requisition if situation so desire. Contractor shall be responsible to take advance action on resource mobilization (men, materials and machine) well in advance to perform shut down works. Adequate manpower shall be mobilise so as to take-up the works in parallel at all supports on given shut-down area. Contractor shall deploy well educated and experience engineer to take care of shut down, and earthing of lines, check clearances on completion of works, return shut down and ensure re-energisation of section/part of line. He shall be available at site for taking shut-down, during execution of works, return of shut down and re-energisation of line. Safety of working crew shall be looked after by him. He must be a trained person having requisite experience of shut-down works. He must be well aware of LT/HT networks and their supply sources.

On completion of work, Contractor shall provide as built GA drawings GPS making of connected consumers and their type of connection (single/three phase), domestic/commercial/agriculture etc. and connected loads.

13.10. Construction of New 11 kV Feeders and Associated Works for Separation of Agricultural Consumers

1. Construction of new 11 kV lines

1.1. Survey

Mapping of routes of proposed 11 KV lines shall be done by GPS survey. While surveying, existing electrical infrastructure, existing agriculture private tube well (PTW) locations, capacity and load details shall be mapped during survey.

Upon completion of the survey the Contractor shall provide a survey report with the following information;

- Single line diagram of the route survey and proposed line alignment details mapped and provided in a standard mapping software. This should also include information such as existing electrical infrastructure and PTW information.
- Road and railway crossing points of the proposed 11 kV line shall be marked on the single line diagram.
- Line sections proposed for 11 kV AB (Aerial Bunched) cables due to vegetation, safety and forest clearance issues (if applicable) shall be marked on the single line diagram. Lengths of proposed AB cable line sections shall be indicated.
- Schedule of location wise pumps to be fed through new feeder or alternatively general consumer distribution transformers to be transferred to new feeder for making dedicated feeder for agricultural consumers. This schedule should include location / capacity of the existing distribution transformers to be transferred and location /capacity and load of agriculture PTWs with reference to the locations of the single line diagram.
- Estimated peak demand on each feeder/s upon implementation and percentage voltage regulation at farthest point on various spur sections.
- Single line diagram of the 33/11 kV substation indicating of 11 kV outgoing feeder to connect the new feeder. If existing spare feeder is unavailable list of works to be performed at the substation to install additional new feeder.

The survey report shall be submitted in a suitable format for uploading to the web portal and for printing.

The survey report will be used as the basic document for execution of work upon approval by the Project Manager. Upon completion of work, as built single line diagram shall be provided with relevant information of the works carried out.

- On the survey drawings Line to line crossing (HT/LT, LT/LT, HT/HT/Railways Crossing) details to be marked with available clearances.
- Efforts shall be taken to avoid long zig-zag lines and too many line-to-line crossing while surveying and finalizing route maps/SLD.

1.2. Statutory Clearances

For execution of new 11 KV line construction work, all statutory clearances shall be obtained by the Contractor coordinating through the Project Manager. These include road crossing clearances, railway crossing approvals, way leave clearance and any approvals needed from local authorities, road authorities and other regulatory authorities. All statutory clearances shall be ensured for ground clearance, line-to-line clearance, road crossing clearance, horizontal and vertical clearances from buildings/objects etc. All road crossings and line crossings shall be guarded as per specifications. Conductor joint should not be provided in mid span length. Instead, it should be nearer to the support.

1.3. Construction of new 11 kV lines

Upon approval from the Project Manager the Contractor shall carry out construction work in full compliance with the technical instructions, specifications and drawings provided in this Volume.

1.4. Stringing of conductors / AB cables

Minimum safety clearances shall be maintained in the 11 kV line. Sag tension charts for installing ACSR conductors or AB cables shall be developed by the Contractor taking into consideration of conductor / cable characteristics, maximum / minimum temperatures and design wind pressure as per service conditions.

Based on this, clear guidelines shall be provided to linesmen to ensure ACSR conductors or bare neutral messenger (for AB cables) are pulled at appropriate tension so that;

- (a) Required ground clearances are maintained and
- (b) Conductor tension is maintained well below its breaking load at all temperatures.

Dynamometer method or sag method may be used to ensure appropriate tensioning of conductors / cables during installation. Over tensioning of conductors should be avoided to ensure its tension does not exceed permissible loading limits at low temperatures. Loose spans of conductors should be avoided to maintain permissible maximum sag at high temperatures. Loose spans may only be allowed for short spans in special cases. This applies when there are practical difficulties to install necessary stays or struts as required at a t-off point. A short loose span may be used in this case to transfer the stay / strut support point upstream or downstream of the line.

Stringing of conductors / AB cable shall be done using proper equipment as recommended by the manufacturer to avoid any damage to the conductor / cable during installation. Temporary stays or strut poles shall be employed as necessary during stringing operation to ensure safety of personnel and equipment.

1.5. 11 kV line for underground railway crossing

Detail survey of location of railway crossing shall be carried out by the Contractor to avoid multi-crossing at nearby location. Prior approval from railway authorities for execution of this work shall be obtained by the Contractor through the Project Manager. Contractor should ensure timely completion of work during block period allocated by the railway authority by mobilizing adequate resources.

2 Nos. separate cables shall be laid in separate HDPE pipe enclosures per crossing. One cable to be kept as a spare. Horizontal Direct Drilling (HDD) shall be used for installation of cables below the railway tracks.

Contractor Contractor Contractor

1.6. Tree Cutting and Trimming

The Contractor shall count, mark with suitable quality of paint all the trees that are required to be cut/trimmed to obtain required wayleave clearance. Contractor shall pay compensation for any loss or damage for tree cutting due to Contractor's work. If forest clearance is envisaged for execution of work, clearance of forest department shall be arranged by the Contractor through the Project Manager.

1.7. Installation of Distribution Box/SMC Distribution Boxes for LT lines

3 phase and single phase Distribution Box/SMC Distribution Boxes shall be supplied as per specifications and installed on poles according to number service connections for LT AB cable lines. When installing single phase Distribution Box/SMC Distribution Boxes they shall be connected to alternate phases of the AB cable in order to obtain adequate level of load balancing.

2. Extension of Substation Control Room Building

Extension of substation control room building shall be carried out as per instructions of the Project Manager to provide additional space for new outgoing feeder panels if required.

This work shall be carried out as instructed by the Project Manager under the supervision of civil engineer / technical officer appointed by him. The Contractor shall be responsible to ensure building is extended in the same manner as existing building. The Contractor shall be responsible to provide necessary measures to ensure substation control operations are carried out without disruption during the construction period.

13.11. Underground Cabling

1. Survey

The detailed survey shall be carried out for the approved feeders/spur lines by the Contractor and submitted for Employer's approval. The Successful Tenderer shall carry out radar survey of the route using Ground penetrating Radar and determine route profile of any other utility cables, pipes etc along the route. The route survey and drilling profile shall be got approved and finalized by the Engineer-in-Charge prior to commencement of the drilling, HDPE pipe insertion and cable insertion.

2. The scope includes :

- Laying of underground 33 KV, XLPE HT Cable (3Cx300 sqmm)
- Laying of underground 11KV, XLPE HT Cable (3Cx185 sqmm)
- Laying of underground 1.1 KV, XLPE, LT Power Cable (3.5Cx240-204.79KM, 3.5Cx 150-227.39KM, 3.5Cx25 sqmm-44.74KM, 1Cx630 Sqmm -1.8KM and 1Cx400 sqmm-2.4KM)
- Laying of underground 1.1 KV, PVC, LT armoured Power Cable for Street Light (4Cx25 sqmm)
- Laying of underground 1.1 KV,PVC, LT armoured Service Power Cable (4Cx10 sqmm - 56KM, 2Cx10 sqmm - 247KM)

3. Horizontal Direction Drilling

- 3.1. HDD or Horizontal Directional Drilling is a trenchless boring method for installing underground cables, pipes and conduits in a shallow curve along a prescribed bore path with the use of a surface-launched rig / machine, which minimises the disruption of the surrounding area, for example, roads and driveways. The laying of U/G cables shall normally be done direct in ground through trenchless boring by using HDPE pipes. However in exceptional circumstances the cables may have to be laid in covered trenches or in racks fixed to the walls or supported from the ceilings. The scope shall cover supply of all the material as per the BOQ, erection equipments, labour and all the other items required for the laying of the power cables. The cable route markers, at a maximum distance of 200 mtrs, and danger boards shall be provided for the information of all concerned and for their safety. Any additional requirement in terms of safety perspective shall be provided by the Contractor without any extra cost. It is the responsibility of the Contractor to maintain the required statutory clearances from other utility services. Any damage caused to any utility services/ human life / public property etc shall be the sole responsibility of the Contractor. The Contractor will lay the underground power cable in such a fashion that no straight through joints are required and only end terminations joints are required, however wherever the joints are required in HT cable then the same shall be carried out overhead on PCC poles structure. In exceptional circumstances such as where length of line is more than the standard cable length in drum and overhead jointing is not possible then straight through joints will be allowed. The Contractor shall have ISO 9001-2008/18001-2007

- 3.2. The Horizontal drilling shall be for a distance of not less than 90 mts at each stretch and subsequently thereafter. The reinstatement of road dug up for drilling at every 90 mts shall be incorporated in the price for Horizontal Directional Drilling.
- 3.3. Disposal of extra excavated material such as mud, slurry, stones etc shall be also included in the rate per meter of horizontal drilling and shall not be charged extra.
- 3.4. The trenchless technology shall be used with HDPE casing for the portion of the cable route such as road, railway, nullah crossing and without HDPE casing for major portion of the cable route. The outer diameter of the HDPE pipe shall be suitable for insertion in an 8" diameter horizontally drilled bore. The HDPE pipe shall be of PE 80 grade with pressure rating PN4 conforming to IS 4984/1995 and shall have wall thickness of 6.20 to 7.10 mm. The HDPE pipes shall be joined by using Butt welding and a 7/20 G.I wire shall be provided along the entire length of each pipe duct.
- 3.5. The directional drilling equipment shall consist of a directional drilling rig of sufficient capacity to perform the bore and pull back the pipe/cable, a drilling fluid mixing, delivery, and recovery system of sufficient capacity to successfully complete the installation, a drilling fluid recycling system to remove solids from the drilling fluid so that the fluid can be reused (if required), a magnetic guidance system or walk over system to accurately guide boring operations, a vacuum truck of sufficient capacity to handle the drilling fluid volume, and trained and competent personnel to operate the system. All equipment shall be in good, safe condition with sufficient supplies, materials and spare parts on hand to maintain the system in good working order for the duration of this project.
- 3.6. The directional drilling machine shall consist of a hydraulically powered system to rotate and push hollow drilling pipe into the ground at a variable angle while delivering a pressurized fluid mixture to a guidable drill (bore) head. The machine shall be anchored to the ground to withstand the bulling, pushing and rotating pressure required to complete the installation. The hydraulic power system shall be self-contained with sufficient pressure and volume to power drilling operations. Hydraulic system shall be free of leaks. Rig shall have a system to monitor and record maximum pullback pressure during pullback operations.
- 3.7. There shall be a system to detect electrical current from the drill string and an audible alarm which automatically sounds when an electrical current is detected.
- 3.8. The drill head shall be steerable by changing its rotation and shall provide necessary cutting surfaces and drilling fluid jets.
- 3.9. Mud motors shall be of adequate power to turn the required drilling tools.
- 3.10. Drill Pipe shall be constructed of high quality 4130 seamless tubing, grade D or better, with threaded box and pins. Tool joints should be hardened to 32-36 RC.

4. Accessories

- 4.1. This being a Turnkey contract, successful installation, commissioning & integration with existing system, of those equipment/accessories/material not specifically mentioned in the specifications, shall be the responsibility of Contractor. No extra payment shall be made for these inherent works.

- 4.2. He shall also supply all other associated equipment/ material/accessories not specifically mentioned in this tender specification but are required for successful and trouble free operation of the executed work as a whole. For that no extra payment shall be made to the Contractor.

5. Technical Standards

- 5.1. The electrical equipments and materials required during erection should be of high standard. Technical features of these equipments and materials must conform to the technical specification given in this bidding document. Wherever the same is not specified, it must conform to the relevant I.S for that material.
- 5.2. Materials conforming to other international standards, which ensure equal or higher quality than the standards mentioned above, shall also be acceptable. In case the bidders who wish to offer materials conforming to other standards, salient points of difference between standards adopted and specific standards shall be clearly brought out in the respective schedule. Four copies of such standards with authentic English version shall be furnished along with the offer.
- 5.3. Whenever a material or an article is specified or described by the name of a particular brand, manufacturer or trade mark, the specific item shall be understood as establishing type, function and quality desired. Products of other manufacturers may also be considered, provided sufficient information is furnished, so as to enable the Employer to determine that the products are equivalent to those mentioned.
- 5.4. Materials supplied/used shall conform in all respects to the relevant Indian Standard Specification with latest amendments there to.

	Title	IS No.
1.	Cement	IS 269
2.	Steel	IS 6003/1970
3.	Fasteners	IS 6639/1972
4.	Concrete mix	IS 1343
5.	RCC	IS 456
6.	Cable laying and jointing	IS 1255

Installation work pertaining to equipment, cable laying etc should be in accordance with the applicable standards, safety codes etc.

6. Site Storage/ Transportation

- 6.1. It shall be the responsibility of the Contractor to store, move/transport from stores/storage yard etc., relevant items and accessories to the place of installation wherever necessary he will assemble all parts of equipment. In accordance with the specific installation instructions as directed by Site Engineer.
- 6.2. The stores should be dismantled and site cleared after the work is completed

7. Erection, Testing and Commissioning

- 7.1. All the works covered under the scope of the tender shall be done in accordance with the norms defined by the Employer, unless the same is not specifically defined in the specification or with the provisions of Indian Electricity Rules/Acts/Other Government Rules/Regulations as prevalent at the time of execution of the job/work.
- 7.2. Installation shall be carried out strictly in accordance with the approved drawings Modifications, if any, required to suit site conditions, shall be carried out only with the prior approval of the Site Engineer. All such changes shall be incorporated in "As built" drawings to be furnished by the Contractor.
- 7.3. Responsibility for successful installation of other equipment accessories, purchased but not mentioned specifically above, and their commissioning shall be on Contractor. For all such items the Contractor shall be supplying all material and equipment required to accomplish the job complete in all respect.
- 7.4. Installation work pertaining to equipment, cable laying etc should be in accordance with the applicable standards, safety codes etc.
- 7.5. The Contractors shall themselves be responsible for timely arrangement/ procurement of all the raw materials required for the manufacture of all tendered items by them/ their and / or by their vendors.
- 7.6. While Repairing & Replacing the equipment, if any other equipment gets damaged due to negligent handling of the Contractor the same shall be replaced by the Contractor at his cost to the Employer satisfaction.
- 7.7. He shall be responsible for dismantling of defective equipments, there proper handling and shifting.
- 7.8. Also he shall hand over the old & dismantled equipments/ material to the purchaser's local stores or other sites as per instructions of the purchaser for which no extra payment shall be made.
- 7.9. All charges on account of damages/losses/claims/thefts etc. involved under the conditions laid down above shall be borne by the Contractor. It's cost shall be recovered from his bills /security deposits /other assets.
- 7.10. In order to avoid hazards to personnel moving around, the equipment such as Transformer, Capacitor Banks, Switchgears etc. if required to be kept charged after installation till their commissioning, shall be cordoned off by suitable barriers to prevent accidental injury to personnel moving around.
- 7.11. Where the equipments/ assemblies are supplied in more than one part, the Contractor shall make all necessary mechanical and electrical connections between the sections. The Contractor shall also do necessary adjustment in the alignments required for its proper operation.
- 7.12. Care shall be taken in handling instruments relays and other delicate devices where instruments and relays are supplied separately they shall be mounted only after the associated switch gear/control panels are erected and aligned.
- 7.13. Precaution: The Contractor shall exercise all possible care to avoid damage to public utilities e.g. water/ sewage pipes telephone and power lines/cable already existing. In case of any accidental

damage during the work, the Contractor shall be responsible to repair/replace the same at his own cost, and shall ensure that the purchaser is not put to any loss.

- 7.14. The Contractor shall have to provide proper lighting, barricading, signboards etc. at the work site as a necessary precautionary arrangement to avoid accident/ damage/ losses to the public /utilities/properties.
- 7.15. Site Solution: It may be possible due to some reasons or others that it would not be possible to work as per the procedure. In such case/cases, the solution to the problem shall be achieved by the purchaser with the consultation of Contractor, and the Contractor shall work as per procedure proposed by the purchaser. Such cases shall in variably be informed to the engineer of the contract for which no extra payments shall be made.
- 7.16. Space Constraints: While executing the job it is quite possible that some of the specified work may not be carried out due to space/land/ other technical constraints etc. In such case the concerned Employer, if required, may divert this work at some other site or cancel the left over portions of work.
- 7.17. The Contractor shall ensure that the equipment under erection as well as the work area and the site are kept clean to the satisfaction of the Engineer. In case, the Engineer is not satisfied about the cleanliness he will have the right to carry out the cleaning operations and expenditure incurred in this regard will be to Contractor's account. Packing cases and packing materials shall be promptly cleared from sites.

13.13. Plinth Mounted Distribution Substations

1.0 SCOPE

The specification covers the design, engineering, manufacture, stage inspection, testing, pre-delivery inspection, supply, delivery, loading, unloading and performance requirements of 11/0.433 KV distribution transformers for outdoor use. The transformers shall be as per latest MSEDCL Technical Specifications.

The equipment offered should have been successfully type tested within five years from date of tender and the designs should have been in satisfactory operation for a period not less than three years as on the date of bid opening. Compliance shall be demonstrated by submitting with the bid, (i) authenticated copies of the type test reports and (ii) performance certificates from the users, specifically from Central Govt./ State Govt. or their undertakings.

The scope of supply should also include the provision of type tests on random samples if desired by the purchaser. In this case the bidder has to bear the charges for conducting such type tests at CPRI or National Govt. approved Laboratory.

The transformer shall conform in all respects to highest standards of engineering, design, workmanship, this specification and the latest revisions of relevant standards at the time of offer and the Purchaser shall have the power to reject any work or material, which, in his judgment, is not in full accordance therewith.

2.0 CODES & STANDARDS

Except where modified by this specification, the transformers shall be designed, manufactured and tested in accordance with the latest editions of the following standards. The Bidder may propose alternative standards, provided it is demonstrated that they give a degree of quality and performance equivalent to or better than the referenced standards. Whether to accept or reject any alternative standard shall be adjudged by the Purchaser. The Bidder shall furnish a copy of the alternative standard proposed along with the bid. If the alternative standard is in a language other than English, an English translation shall be submitted with the standard. In the case of conflict the order of precedence shall be 1) IEC or ISO Standards, 2) Indian Standards, 3) other alternative standards.

IEC/ISO	Indian Standard	Subject
IEC 71		Insulation Coordination
IEC 76	IS 2026	Power Transformers

	IS 1180	Outdoor Three Phase Distribution Transformers up to 500 KVA, 11/ 0.433 KV, Non- Sealed Type
IEC 137	IS 2099	Bushing for Alternating Voltages above 1000V
IEC 156		Method of determining Electric Strength of Insulating Oils.
IEC 296	IS 335	Specification for Unused Mineral Insulating Oils for Transformers and Switchgear
	IS 6792	Method of determination of electric strength of insulating oils.
IEC 354	IS 6600	Loading Guide for oil immersed Transformers
IEC 437		Radio Influence Voltage Measurement
IEC 551		Determination of Transformer and Reactor Sound Levels.
IEC 616		Terminal and Tapping markings for power transformers.
IEC 722		Guide to the Lightning and Switching impulse testing of Power Transformers and Reactors
ISO 1460/BS 729		Galvanizing

This list is not to be considered exhaustive and reference to a particular standard or recommendation in this specification does not relieve the Supplier of the necessity of providing the goods complying with other relevant standards or recommendations.

3.0 SERVICE CONDITIONS

The service conditions shall be as follows:

- maximum altitude above sea level - 1,000m
- maximum ambient air temperature - 50° C
- maximum daily average ambient air temperature - 35° C
- minimum ambient air temperature - 5° C
- maximum temperature attainable by an object exposed to the sun - 60 ° C
- maximum yearly weighted average ambient temperature - 32° C
- maximum relative humidity - 100%
- average number of thunderstorm days per annum (isokeraunic level) - 70
- average number of rainy days per annum - 120
- average annual rainfall - 1500 mm
- maximum wind pressure - 260Kg / m2

Environmentally, the region where the equipment will be installed includes coastal areas, subject to high relative humidity, which can give rise to condensation. Onshore winds will frequently be salt

laden. On occasions, the combination of salt and condensation may create pollution conditions for outdoor insulators.

Therefore, outdoor material and equipment shall be designed and protected for use in exposed, heavily polluted, salty, corrosive, tropical and humid coastal atmosphere.

4.0 SYSTEM CONDITIONS:

The equipment shall be suitable for installation in supply systems of the following characteristics.

Frequency		- 50 Hz $\pm$ 5%
Nominal system voltages	11 KV System	- 11 KV
	LV System	- 433/250 V
Maximum system voltages	11 KV System	- 12 KV
	LV System	- 476 V
Minimum LV voltage	(NEC)	- 392 V
Nominal short circuit apparent power of 11 KV System		- 500 MVA (IS: 2026)
Insulation levels		
1.2/50 μ sec impulse withstand	11 KV System	- 75KV peak (IS:2026)
Power frequency one minute withstand (wet and dry)	11 KV System	- 28 KV (rms)
	LV System	- 3 KV (rms)
Neutral earthing arrangements:	LV System	- Solidly earthed

PART 2: TECHNICAL

		SPECIFIC TECHNICAL REQUIREMENTS	
1		Rated KVA (ONAN rating)	500 KVA, 11/0.433 KV
2		No. of phases	3
3		Type of installation	Outdoor
4		Frequency	50 Hz ($\pm$ 5%)
5		Cooling medium	Insulating Oil (ONAN)
6		Type of mounting	for 500 KVA on Wheels, Mounted on rails.
7		Rated voltage	
	a)	High voltage winding	11 KV
	b)	Low voltage winding	0.433 KV

8		Highest continuous system voltage	
	a)	Maximum system voltage ratio (HV/ LV)	12 KV / 0.476 KV
	b)	Rated voltage ratio (HV / LV)	11 KV / 0.433 KV
9		No. of windings	Two winding Transformers
10		Type of cooling	ONAN (Oil natural / Air natural)
11		KVA Rating corresponding to ONAN cooling system	100%
12		Method of connection:	
		HV:	Delta
		LV:	Star
13		Connection symbol	DYN 11
14		System earthing	Neutral of LV side to be solidly earthed.
15		Percentage impedance voltage on normal tap and KVA base at 750 C corresponding to HV/ LV rating and applicable tolerances (Negative tolerance will not be allowed):	% Impedance + Tolerance %
			5.0 + 10% (No Negative Tolerance)
16		Intended regular cyclic overloading of windings	As per IEC –76-1, Clause 4.2
17	a)	Anticipated unbalanced loading	Around 10%
	b)	Anticipated continuous loading of windings (HV / LV)	110 % of rated current
18	a)	Type of tap changer	Off-load tap changer
	b)	Range of tapping	+ 2.5% to – 7.5% in 5 equal steps of 2.5% each on HV winding
19		Neutral terminal to be brought out	On LV side only

20		Over Voltage operating capability and duration	112.5 % of rated voltage (continuous)
21		Maximum Flux Density in any part of the core and yoke at rated KVA, rated voltage i.e. 11 KV / 0.433 KV and system frequency of 50 HZ	1.5 Tesla
22		Insulation levels for windings :-	
	a)	1.2 / 50 microsecond wave shape Impulse withstand (KVP)	HV: 28 LV: N.A.
	b)	Power frequency voltage withstand (KV- rms)	HV: 28 LV: 03
23		Type of winding insulation	
	a)	HV winding	Uniform
	b)	LV winding	Uniform
24		Withstand time for three phase short circuit	2 Seconds
25		Noise level at rated voltage and frequency	As per NEMA Publication No. TR-1
26		Permissible Temperature Rise over ambient temperature of 50 ⁰ C	
	a)	Of top oil measured by thermometer.	35 ⁰ C
	b)	Of winding measured by resistance.	40 ⁰ C
27		Minimum HV clearances in air (mm) :-	
	a)	Phase to Phase	280
	b)	Phase to ground	140
28		Terminals	
	a)	HV winding line end	12 KV oil filled porcelain communicating type of bushings (Antifog type)

	b)	LV winding	0.433 KV porcelain type of bushings (Antifog type)
29		Insulation level of bushing	HV LV
	a)	Lightning Impulse withstand (KVP)	75 Not applicable
	b)	1 Minute Power Frequency withstand voltage (KV –rms)	28 3
	c)	Creepage distance (mm) (minimum)	25 mm/ KV
30		Material of HV & LV Conductor	Electrolytic Copper for 500 KVA
31		Maximum current density for HV and LV winding for rated current	1.4 Amp/ mm ² for Aluminum windings and 2.4 Amp/ mm ² for Copper windings.
32		Polarisation index i.e ratio of megger values at 600 sec. to 60 sec for HV to earth, L.V to earth and HV to LV.	Shall be greater than or equal to 1.5, but less than or equal to '5'.
33		Core Assembly	Boltless type
34		Maximum permissible No Load and Load Losses (Watts)	500 KVA
	a)	No Load Losses at rated voltage and rated frequency	- 950
	b)	Load Losses at rated current and at 750 C	- 6500

5.0 TYPE OF TRANSFORMER

The transformers shall be of core type construction, double wound, three phase, oil immersed, 11/0.433KV, 50 Hz with natural oil and air cooling (ONAN) to be used as step down transformers for outdoor use. The design of the tank, fittings, bushings, etc shall be such that it will not be necessary to keep the transformer energized to prevent deterioration as the transformers may be held in reserve, outdoors, for many years.

6.0 PERFORMANCE , CAPACITY AND SHORT CIRCUIT RATINGS

The following ratings are covered under this specification

- 500 KVA, 11/0.433 KV, Copper wound

The transformer shall be capable of supplying a continuous load equal to its KVA rating, under the following conditions:

- continuous steady load;
- design at maximum ambient air temperature of 50⁰C;
- 40⁰ C average winding temperature rise and 35⁰C top oil temperature rise for conventional breathing transformers.

The transformer may be overloaded during emergency up to 150% of its continuous rating in accordance with IEC Publication 354 or IS: 6600. Bushings and other current-carrying parts shall also be designed for this condition.

The transformer shall be capable of withstanding for two seconds without damage to any external short circuit, with the short circuit MVA available at the terminals of either winding with rated voltage on the other winding. If short circuit tests have been carried out on the particular design of transformer offered, the test results shall be supplied with the bid.

The thermal ability to withstand short circuit shall be demonstrated by calculation.

The transformer shall be capable of withstanding the thermal and dynamic effects of short circuits, as specified in IEC 76-5 or IS: 2026: Ability to withstand short circuits.

The maximum flux density in any part of the core and yoke at rated KVA, Voltage and frequency shall not exceed 1.5 Tesla.

7.0 VOLTAGE RATIO & TAPPING RANGE

The transformers shall have the following ratio:-

- the nominal voltage ratio shall be 11,000/ 433 V for 500 KVA transformers;
- Tolerance on the voltage ratio shall be $\pm 0.5\%$.

- For transformers of ratings 500 KVA, 11/0.433 KV, Taps shall be provided on the H.V. Windings for voltage variation from tap-1 (+2.5%) to tap-5 (- 7.5%) in equal steps of 2.5 %. Tap No-2 shall be the principal (normal) tap.

The bidder shall state in the technical schedule, the percentage regulation at full load, power factor 1.0 and at full load, power factor 0.8 lagging.

Transformers shall be suitable for parallel operation with each other.

8.0 PERCENTAGE IMPEDANCE

The Percentage of Impedance at 75°C shall be 5% for 500 KVA transformer with positive tolerance of 10%. No negative tolerance on percentage Impedance is allowed.

9.0 LOSSES

The No Load and Load Losses shall not exceed the values given below:-

KVA Rating	Maximum No Load loss in Watts	Maximum Full Load loss in Watts at 75°C.
500 KVA	950	6500

The above losses are maximum allowable and there should not be any positive tolerance.

The offered transformer(s) should have been type-tested at CPRI/ National Govt. approved laboratory. The bid shall be accompanied with type-test reports (short circuit test and Impulse test) conducted at Central Power Research Institute / National Govt. approved laboratory for the offered transformers within five years from date of tender. The short circuit test report(s) must contain the measured no load loss and load loss, determined by CPRI/ National Govt. approved laboratory.

In case of any doubts, <EMPLOYER> reserves the right to verify the original type test reports of CPRI/ National Govt. approved Laboratory or ask the supplier to conduct the type tests at CPRI/ National Govt. approved Laboratory at his (supplier's) cost for re-confirmation of the test results particularly no load losses, load losses and percentage impedance. Bids without type test reports shall not be considered for evaluation.

If the bidder quotes lower values of losses than the CPRI's measured losses, he has to prove the same by conducting the Impulse & short Circuit tests at CPRI/ National Govt. approved laboratory along with measurement of no load losses and load losses at his own cost in presence of EMPLOYER's authorized representative without any financial liability to EMPLOYER.

However, if the loss figures will exceed the stipulated values as per specification, the transformer(s) shall be out rightly rejected.

10.0 VECTOR GROUP

The transformers shall be connected delta-star, in accordance with vector group reference Dyn11 of IEC - 76/ IS - 2026.

The LV neutral shall be brought out to a terminal bushing, which shall be identical to the phase bushings in all respects.

11.0 LOSSES AND CAPITALISATION

Transformers would be out rightly rejected if losses exceed the values indicated at clause-10 above.

12.0 FLUX DENSITY

The flux density at rated voltage & rated frequency shall not exceed. 1.50 Tesla. The transformer must be capable of operating at 10% over voltage and at frequency of 48.5 Hz without saturation.

13.0 INSULATION LEVELS

The insulation levels as defined in IEC 76-C/ IS: 2026 Insulation levels and dielectric test shall apply as per Table.

	HV Winding	LV Winding
Basic Impulse voltage Level (KVp) (1.2/50 micro. sec. Wave)	75	Not Applicable
Power Frequency voltage withstand level, Wet and Dry (KV)	28	3

14.0 NOISE LEVEL

The average noise level of the transformers shall not exceed 51db. The measurement shall be carried out in accordance with IEC 551 at a distance of 300mm from the envelope of the transformer.

15.0 RADIO INFLUENCE VOLTAGE

The maximum radio influence voltage shall be 250 μ V, measured as specified in IEC 437.

16.0 CORE AND WINDINGS

Core

- Stage level inspection for core construction shall be carried out by the Employer.
- Each lamination shall be insulated such that it will not deteriorate due to mechanical pressure and the action of hot transformer oil.
- The core shall be constructed from high grade, non-ageing, Cold Rolled Grain Oriented silicon steel laminations (CRGO or M3 or better) only. No other core materials shall be entertained. Bidders are requested to note that only PRIME CORE materials are to be used. In no case, second grade core material is to be used. The bids should contain copies of invoices towards purchase of core laminations along with test certificates and curves of specific core loss of the laminations proposed to be used.

The purchaser at his discretion, may select samples from the core laminations and get the same tested in CPRI/ Approved National Govt. Laboratory to prove the quality of the core material.

- For the above purpose, the supplier shall have to offer every batch of core laminations received from his Sub-Vendor along with Invoice of the sub-vendor, Mills test certificate, packing list, Bill of lading, Bill of entry certificate to customs etc. towards proof of prime core materials for verification by the Purchaser's representative without any cost to the Purchaser. Besides, the Contractor must mention in his bid about the type of CRGO lamination to be utilized for the offered transformers along with a copy of the specific core loss curve at different flux densities.
- Core materials should be directly procured either from the manufacturer or through their accredited marketing organization of repute, but not through any agent. The core and winding shall be capable of withstanding shocks during transport, installation and service. Provision shall be made to prevent movement of the core and windings relative to the tank during these conditions and also during short circuits.
- The design shall avoid the presence of pockets which would prevent the complete emptying of the tank through the drain valve. The core material offered in the tender to be checked for its correctness before core coil assembly. For this, the tenderer must ask for core and coil inspection before its tanking.
- The laminations shall be free of all burrs and sharp projections. Each sheet shall have an insulating coating resistant to the action of hot oil.
- The insulation structure for the core to bolts and core to clamp plates, shall be such as to withstand 2000 V DC voltage for one minute.
- The completed core and coil shall be so assembled that the axis and the plane of the outer surface of the core assemble shall not deviate from the vertical plane by more than 25mm.
- All steel sections used for supporting the core shall be thoroughly shot or sand blasted, after cutting, drilling and welding.
- The finally assembled core with all the clamping structures shall be free from deformation and shall not vibrate during operation.
- The core clamping structure shall be designed to minimize eddy current loss.
- The framework and clamping arrangements shall be securely earthed.

- The core shall be carefully assembled and rigidly clamped to ensure adequate mechanical strength.
- Oil ducts shall be provided, where necessary, to ensure adequate cooling inside the core. The welding structure and major insulation shall not obstruct the free flow of oil through such ducts.
- The design of magnetic circuit shall be such as to avoid static discharges, development of short circuit paths within itself or to the earth clamping structure and production of flux component at right angle to the plane of the lamination, which may cause local heating. The supporting framework of the cores shall be so designed as to avoid the presence of pockets, which would prevent complete emptying of the tank through the drain valve or cause trapping of air during filling.
- The construction is to be of boltless core type. The core shall be provided with lugs suitable for lifting the complete core and coil assembly. The core and coil assembly shall be so fixed in the tank that shifting will not occur during transport or short circuits.

INTERNAL EARTHING

- All internal metal parts of the transformer, with the exception of individual laminations and their individual clamping plates shall be earthed.
- The top clamping structure shall be connected to the tank by a copper strap. The bottom clamping structure shall be earthed by one or more the following methods:
 - a) By connection through vertical tie-rods to the top structure.
 - b) By direct metal to metal contact with the tank base.
 - c) By a connection to the structure on the same side of the core as the main earth connection to the tank.
- The magnetic circuit shall be connected to the clamping structure at one point only and this shall be brought out of the top cover of the transformer tank through a suitably rated insulator. A disconnecting link shall be provided on transformer tank to facilitate disconnections from ground for IR measurement purpose.
- Coil clamping rings of metal at earth potential shall be connected to the adjacent core clamping structure on the same side as the main earth connections.

Windings

- Winding shall be subjected to a shrinking and seasoning process, so that no further shrinkage occurs during service. Adjustable devices shall be provided for taking up possible shrinkage in service.
- All low voltage windings for use in the circular coil concentric winding shall be wound on a performed insulating cylinder for mechanical protection of the winding in handling and placing around the core.

- Winding shall not contain sharp bends which might damage the insulation or produce high dielectric stresses. No strip conductor wound on edge shall have width exceeding six times the thickness.
- The winding insulation shall be free from insulating compounds which are liable to soften, ooze out, shrink or collapse. It shall be non catalytic and chemically inert in hot transformer oil during normal service.
- The stacks of windings are to receive adequate shrinkage treatment.
- The windings and connections are to be braced to withstand shocks during transport, switching, short circuit or other transient conditions.
- Permanent current carrying joints in the windings and leads shall be welded or brazed. Clamping bolts for current carrying parts inside oil shall be made of oil resistant material which shall not be affected by acidity in the oil steel bolts, if used, shall be suitably treated.
- Terminals of all windings shall be brought out of the tank through bushings for external connections.
- The windings shall be uniformly insulated and the L.V neutral points shall be insulated for full voltage.
- The completed core and coil assemble shall be dried in vacuum at not more than 0.5mm of mercury absolute pressure and shall be immediately impregnated with oil after the drying process to ensure the elimination of air and moisture within the insulation. Vacuum may be applied in either vacuum over or in the transformer tank.
- The winding shall be so designed that all coil assemblies of identical voltage ratings shall be interchangeable and field repairs to the winding can be made readily without special equipment. The coils shall have high dielectric strength.
- Coils shall be made of continuous smooth high grade electrolytic copper or aluminium conductor shaped and braced to provide for expansion and contraction due to temperature changes.
- Adequate barriers shall be provided between coils and core and between high and low voltage coil. End turn shall have additional protection against abnormal line disturbances
- The insulation of winding shall be designed to withstand voltage stress arising from surge in transmission lines due to atmospheric or transient conditions caused by switching etc
- Tapping shall not be brought out from inside the coil or from intermediate turns and shall be so arranged as to preserve as far as possible magnetic balance of transformer at all voltage ratios.
- Magnitude of impulse surges transferred from HV to LV windings by electromagnetic induction and capacitance coupling shall be limited to BIL of LV winding.
- The winding conductor shall be of Al. up to 250 KVA transformers and copper for 500 KVA Transformers. The current density shall not exceed 1.4 Amp/ mm² for aluminium and 2.4 Amp/ mm² for copper at normal full load current.

17.0 BUSHINGS AND TERMINATIONS